CSC 3320: System-Level Programming

Review for Exam 2

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Note: These slides are updated based on original notes from Raj Sunderraman, Michael Weeks, and Yuan Long

Announcements

- Exam 2 date (March 26, 2026) will be announced on iCollege
  - Syllabus:
    - Chapter 3: I/O
      Chapter 4: Expressions
      Chapter 5: Selection Statements
      Chapter 6: Loops
    - Chapter 8: Arrays
      Chapter 9: Functions
    - Chapter 11: Pointers
    - Chapter 12: Pointers and Arrays
- One A4 page, double-sided, hand-written, cheat sheet is allowed.

Disclaimer

This review merely highlights some of the important areas that you may focus for exam 2. Samples shown in this review are just few examples (not exhaustive in terms of either examples or topics). For topics, refer to the broader list I have provided as Exam 2 guideline. For complete preparation, please refer to all weekly slides/books/lecture notes for the chapters mentioned in the previous slide.

Question Types and Distributions

- MCQ (~40%)
  - May involve coding questions (specifically output determination)
  - Other conceptual questions
- Short Answer type (40-50%)
  - Piece of code writing, output determination w/o options, comprehension questions
- Write/Fill in small piece of code or small programs. (10-20%)
  - Fill-in codes
  - Write small piece of programs

Formatted Input – An example

- Both `scanf("%d %d", &a, &b);` and `scanf("%d%d", &a, &b);` will function identically in practice.

- However, spaces work differently for characters. Spaces in the format specifier such as `("%c %c")` skip whitespace between characters, while the version without spaces `("%c%c")` does not, reading all characters as they appear in the input.

- Understand many other variations. We have seen 1 example in hw3.

An example assignment

```
int x = 5, y = 3, z;
z = (x < y);
```

What will be the value of z?

Increment/Decrement Operators

```
int main ( )
{
    int a, b, c, d;
    a=15; b=10;
    c =++a - b;
    d=b++ +a;
}
```

What will be the value of `a, b, c,` and `d`?

Increment/Decrement ops

What will be the value of x and y after the execution of the following statements?

```
int i=6;
int j=4;
int y;
y=(j++)*(++i);
```

What will be the value of i, j, and y?

Continue/break statement

```
int main( )
{
    int i, j, sum=0;
    for(i=1; i<=10; ++i)
    {
        if(i%2==0)
            continue;
        for(j=1; j<=i; j++)
        {
            sum+=j;
            break;
        }
    }
}
```

What will be the value of `sum`?

ASCII values

What will be the value of x after execution of the following statements?

```
int x, y=10;
char z='2';
x=y+z;
```

switch vs. else-if ladder

```
int main( )
{
    int a=5;
    switch(a)
    {
            case 5: printf("Five");
            case 6: printf("Six");
            case 7: printf("Seven");
            default: printf("Not Found");

    }
}
```

What will be the output?

Conditional Operators

```
int n = 49;
n % 3 == 0 ? printf(“Hello!") : printf(“World");
```

What will be the output?

Logical operators

```
#include <stdio.h>
int main()
{
    int a = 0;
    int b;
    a = (a == (a == 1));
    printf("%d", a);
    return 0;
}
```

What will be the output?

Short-circuited logical operation

```
#include <stdio.h>

int main()
{
   int x = 5;
   int y = 0;
   int z = 1 || --x;
   int d = 0 && --x;
   printf(”x = %d, y = %d, z = %d, d = %d", x, y, z, d);
   return 0;
}

Output: x = 5, y = 0, z = 1, d = 0
```

Static and Global Variables

```c
#include <stdio.h>

int x = 10;

void change() {
    static int x = 5;
    x++;
    printf("%d ", x);
}

int main() {
    change();
    change();
    printf("%d", x);
    return 0;
}
```

Pointers

```
int main( )
{
   int m=10, n, *b;
   b=&m;
   n=*b;
   m=*b+n;
}
```

What will be the value of m and n?

Call by Address/Value

```c
#include <stdio.h>
void increment(int *n) {
    (*n)++;
}

int main() {
    int a = 10;
    increment(&a);
    printf("%d", a);
}
```

Answer: 11

```c
#include <stdio.h>
void increment(int n) {
    n++;
}

int main() {
    int a = 10;
    increment(a);
    printf("%d", a);
}
```

Answer: 10

Array Manipulation

```
int main( )
{
    int i, DATA[10]={ 1, 2, 3, 4, 5, 6, 7, 8, 9, 10};
    for(i=0; i<10; ++i)
    {
        if(i%2==0)
            DATA[i]=2*DATA[i];
        else

            DATA[i]=2*DATA[i]-1;
    }
    for(i=1; i<=10; i++)
        printf("%4d", DATA[i]);
}
```

What will be printed on the screen?

Arrays and Pointers

Q. What is the relationship between an array name and a pointer in C?

A. The array name is a constant pointer to the first element.

B. The array name stores the address of the last element.

C. The array name is a variable pointer.

D. The array name cannot be used as a pointer.

Arrays and Pointers

What will be the output?

```
#include <stdio.h>
void fun(int *ptr) {
    *(ptr + 1) = 99;
}

int main() {
    int arr[3] = {10, 20, 30};
    fun(arr);
    printf("%d", arr[1]);
    return 0;
}
```

Answer: 99

Reminder

- This is just a few examples. Go back to the topics list and read from weekly slides, notes, HW3 and HW4, and other sources.